

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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Anonymous submission

ABSTRACT

We explore how physical scale and population size shape the emergence of complex behaviors across open-ended ecological environments. In these settings, agents are unsupervised and have no explicit rewards or learning objectives but instead evolve over time according to reproduction, mutation, and selection. As they act, agents continually reshape their environment and the population around them in an ongoing, dynamic ecology. Our goal is not to optimize a single high-performance policy, but rather to characterize how broad families of behaviors emerge and evolve across large populations under natural competition and environmental pressure. Using modern hardware together with a new multi-agent simulator, we scale both the environment and the population well beyond the sizes previously attempted, reaching populations of over 60,000 agents, each carrying its own evolved neural network policy. Across these settings we identify a range of emergent behaviors—including long-range resource extraction, vision-based foraging, and predation—that arise generically under competitive and survival pressures. Examining how sensing modalities and environmental scale govern the emergence of these behaviors, we find that some of them appear only once the environment and population grow sufficiently large, and that increasing scale broadly improves the stability and consistency of the behaviors that emerge. While there is a rich history of research in evolutionary settings, our scaling results on modern hardware point to broadly applicable new directions for treating ecology as an instrument of machine learning in an era of increasingly abundant computational resources and efficient machine frameworks. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been studied across a wide variety of biological disciplines, spanning evolutionary theory (Darwin 1859, Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). Models of these emergent dynamics range broadly, from theories of speciation and adaptive radiation (Coyne & Orr, 2004, Schluter, 2000) to the mathematical analyses of population genetics (Fisher 1930, Wright 1931).

Across these fields, decades of mathematical modeling, laboratory experiments, and field work have tremendously improved our understanding of the mechanisms of evolution in nature. Yet studying emergent behavior in the natural world remains beset by fundamental obstacles. As ecosystems grow larger and more complex, they become correspondingly harder to control and to measure. Even where such study is feasible, running controlled experiments in large-scale natural settings risks damaging or displacing wild populations.

To better understand how complex behavior emerges under competitive and environmental pressures, we study the open-ended evolution of neural network policies in large-scale ecological simulations. In these settings, agents have no specified objective or reward signal, and instead collect resources to survive and reproduce according to the dynamics of the environment. Policy changes occur only via mutation, as parents pass their policies on to their children. With the rapid growth in computational resources available to researchers, our aim is to study open-ended evolutionary dynamics in populations of such simple digital organisms at scales not previously possible, pro-

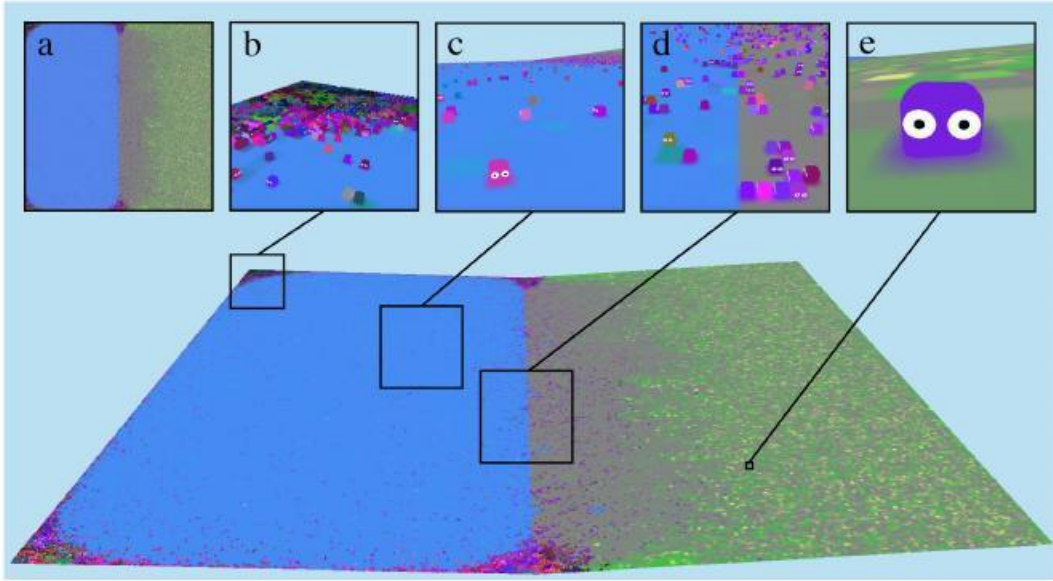


Figure 1: A 3D visualization of a representative 512×512 map in our environment. Insert (a) shows a top-down view of the entire map, (b) shows a group of agents clustered in the corner, (c) shows agents in the water, (d) shows agents travelling back and forth from the land, and (e) shows a close-up of a single agent gathering resources.

viding general-purpose tools for investigating how environmental scale and population size shape behavioral emergence.

To this end, we build a new JAX-based simulation environment that supports rapid evaluation of large grid worlds across a wide range of configurations. Our larger experiments can contain over 60,000 individual agents within a physical area of over 1,000,000 grid cells. Agents in these worlds must navigate to find food and other resources in order to survive, and can reproduce by collecting enough resources to make a mutated copy of themselves. A 3D visualization of this environment can be found in Figure 1.

Our goal is not only to better understand ecological phenomena through simulation, but also to study ecology broadly as a mechanism for producing machine intelligence. Recent years have demonstrated that in AI, scale does not merely improve models—it fundamentally transforms their capabilities (Wei et al., 2022a). Yet despite this insight, embodied learning has largely been confined to relatively small environments with low populations. Our work asks what capabilities emerge when we scale not just a single model but an entire ecosystem. Just as reasoning emerges in individual models only beyond a certain size threshold (Wei et al., 2022b), we seek to understand which classes of strategies evolve within populations only once environmental size and complexity become sufficiently rich.

Through extensive experiments, we identify a range of settings in which sensor configurations and large environmental scales strongly shape the behaviors that emerge from simulation, often with long-term ecological consequences. For instance, we show that agents equipped with simple compass sensors can adapt to conduct long-range resource-gathering expeditions, whereas this behavior emerges far less reliably at small scales. Similarly, agents with vision sensors evolve markedly better foraging and attacking behavior than agents without them, and again these effects appear more reliably as scale increases. Taken together, these experiments indicate that the scale and complexity enabled by modern hardware accelerators offer broadly promising approaches for studying how evolution in ecological environments can produce rich new behaviors.

2 RELATED WORK

The emergence of life has inspired computational research since the field’s inception. Conway’s Game of Life (Games, 1970) shows how populations can arise from nothing more than simple local rules. Early computer programs such as Tierra (Ray, 1992) and Avida (Brown) model life as self-replicating programs that evolve through mutation. In the same spirit, Reynolds (1987) used local rules of motion to produce emergent flocking and schooling behaviors. Across these lines of work, a recurring theme is that rich collective behavior can emerge from minimal underlying rules.

Researchers have applied these principles across a wide range of settings, evolving the locomotion morphology (Sims, 1994a, b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and sensing (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025) of virtual agents, while neuroevolution methods evolve neural network architectures directly (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). In contrast to these approaches, we do not attempt to produce agents that better adapt to their environment through a reward or objective signal. Instead, we seek to characterize which behaviors naturally emerge under ecological pressure, and how environmental scale and complexity shape those behaviors across conditions.

Evolution has also provided inspiration for the internal dynamics of optimization algorithms more broadly. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) apply the principles of evolution and natural selection to optimization across many problem domains. Related methods use evolutionary search to encourage novelty within the resulting solutions (Lehman & Stanley, 2011; Cully et al., 2015), and such approaches have been used to optimize neural networks for tasks ranging from Atari games to Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). While our work likewise employs evolutionary mechanisms to update agents, our focus is on studying emergent behavior in its own right, rather than developing a new optimization algorithm or applying an existing one to accomplish a fixed objective.

There is also a rich history of work that draws on biological inspiration to develop organisms within an environment. Early examples include Bedau (1992), Yeager et al. (1994), Channon & Dampier (1998), and Tisue et al. (2004). More recent efforts use cellular automata to grow organisms with emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025). Mordatch & Abbeel (2018) and Park et al. (2023) use simulated environments to investigate the emergence of language and communication between agents. The emergence of behavior via autocurricula (Leibo et al., 2019)—the idea that populations of agents provide an increasingly challenging learning landscape for one another—has likewise been explored, though in environments with smaller populations than ours (Leibo et al., 2018; Team et al., 2021). Additionally, Wang et al. (2019) and Aki et al. (2024) take the reverse approach, evolving environments specifically optimized for ideal agent learning. Although these works evolve the environment rather than the agent, they are directly relevant to ours in highlighting how environmental complexity can, in general, elicit sophisticated behaviors.

Most closely related to our work is a broader body of efforts that simulate populations of agents interacting within complex environments. Baker et al. (2019) examine a Hide-and-Seek task in which a small number of hider agents learn to use environmental tools to evade seeker agents, both trained via reinforcement learning. Other works similarly train populations for specific tasks, spanning playing video games (Jaderberg et al., 2019), manipulating objects (Petrenko et al., 2021), and capturing agents (Zheng et al., 2017; Yamada et al., 2020). However, while many of these works explore emergence, they generally do so within fixed tasks and objectives. The Neural MMO framework (Suarez et al., 2019; 2023) takes a more open-ended approach to reinforcement learning, placing agents within a massively multi-agent game setting and optimizing only for survival rather than fixed objectives. However, its population remains capped at a fixed number of agents (experiments examine up to 128 concurrent agents) without generational inheritance, limiting the range of population dynamics that can be examined. Hamon et al. (2023) aligns more closely with our work through variable population sizes, genetically inspired evolution, and generational timelines, though its environment is considerably smaller and less feature-rich, and it does not directly explore the role of scale. Similarly, Lu et al. (2024) also simulates agents evolving to gather resources. That work,

too, operates at a much smaller scale (a maximum of 256 agents) and emphasizes tool use through programmable robots. While all of these past works take valuable steps toward examining emergent ecological dynamics, a comprehensive understanding requires populations that can grow and shrink naturally, evolve without predefined objectives, and interact within environments rich enough to support diverse strategies at scales where collective patterns can emerge. Our work brings these elements together, enabling the study of a broad class of ecological phenomena that were inaccessible in prior work.

Finally, we are also inspired by a growing body of work that emphasizes environmental speed and scale, using optimized libraries to increase the computational throughput and the number of agents able to interact in game-like environments (Koyamada et al., 2023; Matthews et al., 2024; Lechner et al., 2023). We build directly on this progress in computational efficiency, adopting similar JAX-based acceleration techniques to achieve large scales and fast simulation across ecological environments.

3 METHOD

3.1 ECOLOGICAL GAMES

We model the interactions between populations of agents and their environment as a general type of multiplayer game we refer to as an **Ecological Game (EG)**. Much like the mathematical models used across sequential decision-making—Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see Sutton et al., 1998 and Albrecht et al., 2024 for reference)—an EG carries underlying Markovian transition dynamics, but is designed to express changing populations of agents in an objective-free setting. Like an SG, an EG contains a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ mapping states s_t and a set of actions for each player $a_{t,i}$ to a distribution over subsequent states s_{t+1} . Unlike an SG, it removes the reward function and adds a population function $\Gamma(s_t)$ that reports which agents in I are currently alive and the identity of their parents, so that birth and death events can be inferred. Just as MDPs and SGs admit partially observable counterparts—POMDPs and POSGs—an EG can also be partially observable, in which case we refer to it as a POEG. A POEG differs from an EG in that it does not allow each agent to observe the entire state, and instead incorporates an observation function $O(s_t, i)$ that maps a state and player to an observation for that player. This formulation is deliberately general; the environment we study is one such POEG, instantiated with the different observation functions corresponding to the sensor configurations described in the next subsection.

3.2 ENVIRONMENT

The particular POEG we instantiate for our experiments is a large grid world containing a variety of resources that agents must collect in order to reproduce and survive. The environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, where s_t^m is 2D data associated with the map, such as the terrain and distribution of resources, and s_t^a is data associated with the agents, such as position, health, and internal resources.

Terrain: Each map has a rock layer defining the height of every grid cell. Maps also include water, which is initially added to grid cells with a rock value below a preset sea-level. Throughout an episode, water flows downhill according to a discrete flow simulation driven by the height defined by this rock layer. This construction supports a broad family of terrains; in the experiments below, we use six representative global configurations shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. We conduct experiments across grid sizes spanning 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 . When growing or shrinking the grid, we scale the initial population size in proportion to the change in overall grid area. Once the simulation begins, the population then fluctuates freely according to the dynamics of the environment.

Resources and Agent Health: To survive and reproduce in these environments, agents must collect **water**, **energy**, and **biomass**. Biomass and energy are initially distributed at random throughout the environment, but can be consumed and redistributed by individual agents.

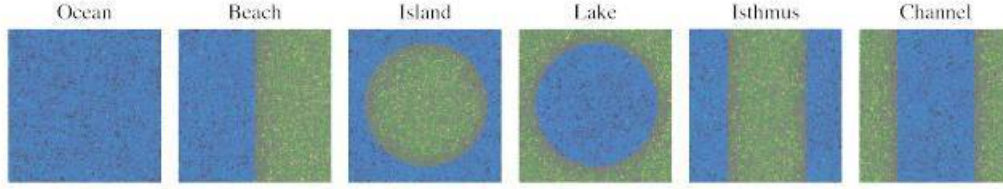


Figure 2: Overhead images of the representative terrains used in our experiments, shown at the 256×256 grid size. Small dark spots represent individual agents, while green and yellow represent concentrations of energy and biomass. Each image was captured 5000 steps into an episode, after most biomass in the water has been eaten but before the agents have evolved to survive on land for longer periods.

Each action an agent takes expends a small amount of energy and water. Spent water returns to the landscape, where it flows downhill to the nearest body of water, while the energy is consumed and destroyed. Biomass is used for reproduction: all agents require a fixed amount of biomass to exist, and therefore cannot reproduce until they have collected enough excess biomass to donate to their offspring. Like water, the total biomass in the environment remains constant throughout the simulation, but unlike water it neither flows downhill nor moves on its own. Energy is not held constant, but instead slowly accumulates on free-standing biomass. We designed these resource dynamics to pose a somewhat more demanding problem than collecting a single food source, and more generally to capture the kinds of coupled resource constraints that arise in richer ecologies. For ablations on resource rules, see Appendix F.2

Agents also carry health points (HP) that they must maintain in order to avoid death. Overspending a critical resource causes an agent’s health to deteriorate rapidly, though it may use energy and water to slowly recover HP over time. Agents may additionally attack one another, reducing the target’s HP, and take a small but increasing amount of damage as they age, so that agents eventually die of old age if not by other causes. When an agent’s HP reaches zero and it dies, whatever resources it was carrying are returned to the environment in the grid cell where it was standing. In this way, an agent that attacks and kills another may then consume its resources, giving rise to a basic form of predation. For ablations on HP hyperparameters, see Appendix F.8

Sensors: Agents sense the environment through four complementary sensors. The **internal** sensor reports the agent’s age, health, and the resources it is carrying; the **external** sensor detects external resources in the same grid cell as the agent; a **compass**, inspired by the magnetoreceptors found in birds and other animals, reports the agent’s global direction as a 4-way one-hot vector; and **vision** provides a 7×7 top-down rendered patch around the agent, with three color channels together with a fourth channel encoding the local difference in elevation. This modular set of sensors lets us study a range of sensing regimes; for ablations on vision configurations, see Appendix F.6

Actions: Agents choose from a finite set of discrete actions grouped into five categories. The first is **Rest**, which does nothing but costs the least energy. There are three **Move** actions: rotate clockwise, rotate counterclockwise, and move forward one grid cell. Multiple agents cannot occupy the same cell; if an agent tries to move into a cell occupied by another agent, or if two agents attempt to move into the same cell, the move action is canceled. Agents also have a single **Attack** action, which kills all agents in the 3×3 square directly in front of them. For ablations on attack strength, see Appendix F.7 There are three separate **Eat** actions, one per resource type. When an agent shares a cell with a particular resource, taking the corresponding eat action transfers a fixed amount of that resource from the environment into the agent’s stomach, which has a fixed maximum capacity.

Finally, agents have a single **Reproduce** action that produces a new offspring via single-parent reproduction. This action succeeds only if the agent has accumulated enough resources to donate to its new child. The offspring is placed directly behind its parent, its policy is a mutated copy of the parent’s weights, and it immediately inherits a fixed amount of the parent’s resources.

Throughout the experiments below, we selectively disable certain sensors or the attack action in order to probe the ecological effects of agents with differing abilities, allowing us to compare across a spectrum of agent capabilities. We consider three agent sensing combinations: internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**).

We write **+A** to indicate when attacking is allowed; for example, **(RC+A)** denotes an agent with resource sensors, a compass, and the ability to attack.

3.3 POLICIES

Our agent policies are implemented as small, memoryless MLPs. We opt for agents without memory in order to keep the computational model simple and to reduce the number of network parameters. The raw values from each sensor are flattened, normalized to the range $[-1, 1]$, and passed through separate linear layers with an output dimension of 64. These are summed and fed through a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final output is a linear layer whose output dimension equals the number of possible actions. Actions are sampled from a softmax distribution whose temperature is itself allowed to evolve alongside the network weights. Weights are stored as half-precision bfloat16 values to support larger population sizes and faster run times. Depending on the number of sensors in use, each agent’s network ranges roughly from 10k to 25k parameters. For ablations on network size and architecture, see Appendix [F.5](#).

In all runs, the weights of the initial population use Kaiming initialization ([He et al., 2015](#)), with bias vectors initialized to zero. Policies for agents born mid-episode are created by copying the parent’s weights and applying Gaussian noise with standard deviation 3×10^{-2} to all mutable weight values. This mutation approach follows common practice in neuroevolution research ([Such et al., 2017](#)). While more elaborate mechanisms such as crossover are certainly possible, we adopt this simple scheme so as to isolate the effect of environmental scale in as clean a setting as possible. Empirically, we found that a larger mutation rate was necessary because of the low-precision floating-point representation; we present an ablation on mutation rates in Appendix [F.4](#).

Agents also carry a 3-channel trait controlling their color when rendered onto the map. This color trait mutates in the same manner as the network weights and determines how an agent appears to other agents equipped with the vision sensor. In principle, this allows agents to distinguish among different types of neighbors.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX ([Bradbury et al., 2018](#)) and run on Nvidia H100 and A40 GPUs using a state-of-the-art simulation environment. Unless otherwise noted, every experiment runs for 2M environment steps or until extinction, with per-seed wall-clock times spanning from under an hour for 64×64 grids to roughly 10 hours for 1024×1024 grids on a single H100 GPU. Appendix [E](#) describes the simulator architecture in more detail, reports additional scaling measurements, and discusses reproducibility considerations.

4 EXPERIMENTS

In the experiments below, we characterize how different sensing modalities, environmental scale, and population size shape the emergent behavior of agents across a range of conditions.

4.1 LONG-DISTANCE RESOURCE GATHERING

In our first set of experiments, we study the emergent ability of agents to travel long distances inland in search of resources. We find that whenever the environment offers clear directional structure, agents equipped with a compass (**RC**) consistently develop far more effective long-range resource-gathering behavior than a baseline agent class (**R**) without a compass. We further show that the emergence of this behavior is more consistent and more effective as environmental scale grows.

We begin with a visual example of the (**R**) agent class, shown in Figure [3](#). After 5000 time steps (a), all the initial agents on dry land have died from lack of water, while those in the water have multiplied rapidly thanks to the rich initial resources. At this early stage, the population does not yet contain sophisticated policies capable of efficiently seeking out resources. From this early peak, agents in the water occasionally stumble onto land (b). When they do, they typically cannot find their way back to the ocean and die, depositing their biomass on land. By 50,000 time steps (c),

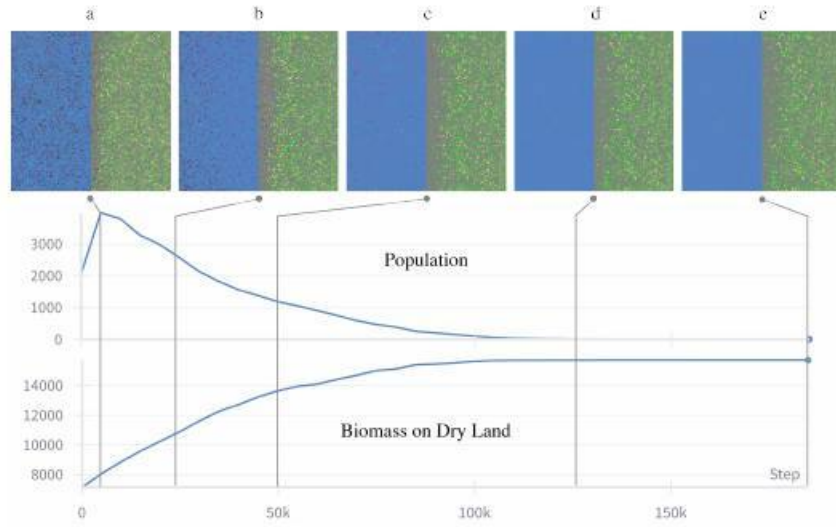


Figure 3: A representative trial with **(R)** agents in a 256×256 **Beach** environment. The top chart shows the population over time, and the bottom chart shows the free biomass on dry land.

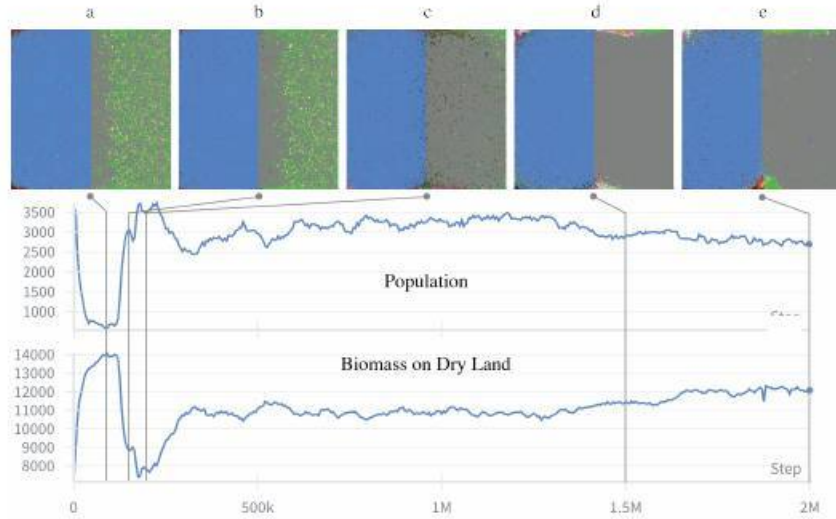


Figure 4: A representative trial with **(RC)** agents in a 256×256 **Beach** environment. The top chart shows the population over time, and the bottom chart shows the free biomass on dry land.

this gradual process has nearly doubled the amount of uneaten biomass on land. This transfer of resources from water to land drives the population down, since there is no longer enough biomass in the water to sustain a large population. By 125,000 steps (d), almost all biomass has been transferred to land and the population is nearly extinct. The final agent dies around 185,000 steps (e).

This example illustrates a general ecological consequence of the agents' inability to find their way back to water. Not only do individuals die, but their deaths reshape the distribution of resources in the environment in ways that carry important consequences for the entire population. At the same time, this ecological catastrophe opens an opportunity for any agent that manages to discover how to walk onto land, forage in the rich deposits of biomass left by its dead ancestors, and safely return to water.

In the next example, shown in Figure 4, we equip agents with an internal compass **(RC)**. At first, a similar pattern emerges (a): the population shrinks while the free biomass on land rapidly increases.

In this case, however, some agents are able to use their compass to drift into the relative safety of the corners. Around 150,000 steps (b), some agents begin to “mine” the biomass on land, using their compass to navigate back and forth between land and water. By 200,000 steps (c) these miners have cleared the entire beach. The population then undergoes only minor changes, with agents forming clumps around the edges (d), and remains largely stable for the full 2,000,000 time steps (e).

Table 1: Mining events in the **Beach** world for populations of agents with a compass (**RC**) compared to populations without a compass (**R**), spanning five world sizes and initial populations.  denotes the number of seeds that featured a mining event, and  denotes the number of seeds in which the population went extinct before 2M steps. See Table 3 in Appendix A for detailed per-seed data.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
										
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

To quantify these observations across scales, we run a full set of **Beach** experiments at grid sizes 1024 × 1024, 512 × 512, 256 × 256, 128 × 128, and 64 × 64, with initial populations of 32768, 8192, 2048, 512, and 128 agents, respectively. Each experiment is run with four random seeds, and we record whether mining behavior emerges, which we define as a drop of at least 10% in the free biomass on dry land from its historical high point. We also record the onset time of the mining event, determined by the local maximum in biomass on land, along with the percentage drop in biomass. A summary of mining and extinction events appears in Table 1 with full per-seed details in Table 3 in Appendix A. Mining behavior emerges consistently in (**RC**) agents but rarely in (**R**) agents. Its emergence becomes steadily more reliable as environments grow, occurring in all seeds from the 256×256 scale upward. At smaller scales, populations become less stable and more prone to early extinction.

To probe these effects more closely, we run additional seeds with (**R**) and (**RC**) agents at the 512×512 world size (32 seeds) and the 128×128 world size (64 seeds), and use a Kaplan-Meier estimator to measure the probability of extinction and of mining as a function of simulator time. Extinction becomes rarer and slower to occur as world size grows, while at smaller grid sizes mining is more frequent and arrives sooner. We also fit a Cox proportional hazards model comparing the timing of mining discovery and extinction across world sizes and compass usage, and find consistently that larger worlds substantially reduce the rate of extinction and enable much more rapid mining discovery than smaller worlds. Enabling the compass produces a comparable effect. Full details can be found in Appendix B.

Table 2: Mining and extinction events across the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for populations of agents with a compass (**RC**) compared to those without a compass (**R**), at the 512 × 512 grid size.  denotes the number of seeds that featured a mining event, and  denotes the number of seeds in which the population went extinct before 2M steps. See Table 4 in Appendix A for detailed per-seed data.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

To assess how broadly these findings hold across terrains, we run analogous experiments in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at the 512 × 512 grid size with (**R**) and (**RC**) agents, examining how terrain configuration shapes the emergence of mining behavior. Mining and extinction events for these experiments appear in Table 2 with detailed per-run information in Table 4 in Appendix A. As before, equipping agents with a compass yields earlier and more consistent

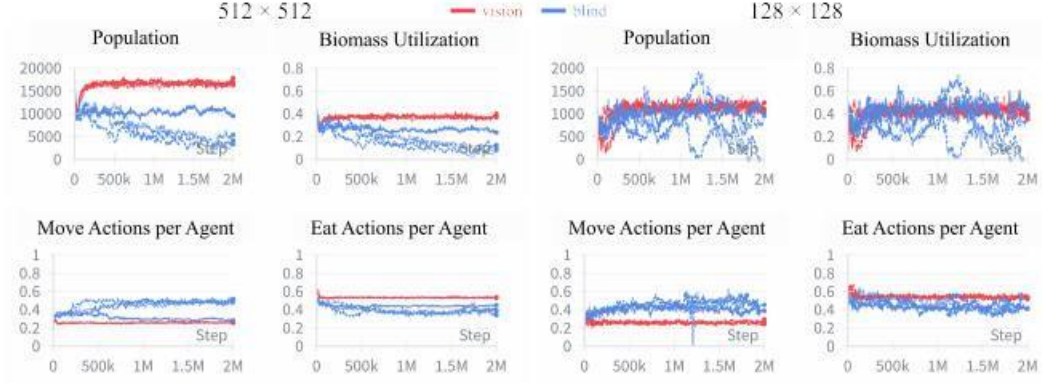


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in the 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agent populations are larger and more stable, achieving greater biomass utilization through less moving and more frequent, more effective eating. At larger scales these dynamics grow more stable, with less variance across runs, suggesting that as environments grow the vision sensor offers an increasing advantage over blindness for adapting foraging behavior. See Figure 9 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

emergence of mining behavior, along with more stable populations and lower extinction rates. (**RC**) agents adapt more readily to the island and isthmus environments than to the lake and channel, where they occasionally go extinct. We attribute this to a simple directional cue: walking straight east or west always returns an agent to water in the island and isthmus, whereas no such easy rule exists in the channel and lake. The (**R**) agent cannot reliably adapt to any of these environments, though it struggles least on the island. See Figure 12 in Appendix D for a visual of mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

We next turn to the **Ocean** environment to determine whether agents can adapt to use vision sensors to effectively locate free resources or prey on other agents. We compare (**RCV+A**) agents, which have resource sensing, a compass, and a 7×7 visual sensor, against (**RC+A**) agents with only resource and compass sensors.

Figure 5 plots population, biomass utilization, movement actions, and eat actions for agents in the 512×512 and 128×128 grid sizes. Plots for other grid sizes appear in Figure 9 in Appendix C and summary statistics of these plots appear in Table 5 in Appendix A. Biomass utilization is computed as the percentage of the total biomass in the environment that is currently held inside the agents. Whenever agents have vision sensors (red lines), they consistently spend less time moving and more time eating. These runs also support larger populations and achieve greater biomass utilization. Together, these dynamics indicate that the agents have adapted to use their visual sensing to forage for food more efficiently. Comparing across environmental scales, the variance between runs decreases substantially as environments grow. Notably, runs that include vision sensors exhibit much more stable dynamics, with less variation in these metrics over time.

Figure 6 shows the averaged attacks per agent and homicides per attack for these runs. Agents with vision attack relatively infrequently yet maintain a successful kill rate of approximately 75%. In large grids, agents without vision sometimes also achieve high kill rates, though with much more unstable dynamics. Several confounding factors make the underlying causes of this observation difficult to pin down. Through visual inspection (see Figure 13 in Appendix D), we find that these high kill rates sometimes arise when clusters of agents build up in one location, and that such clusters almost never form in runs with visual agents, which may further indicate that agents with vision can adapt to avoid one another. In smaller environments the gap between visual and blind agents is less pronounced, though populations of visual agents remain far more stable throughout.

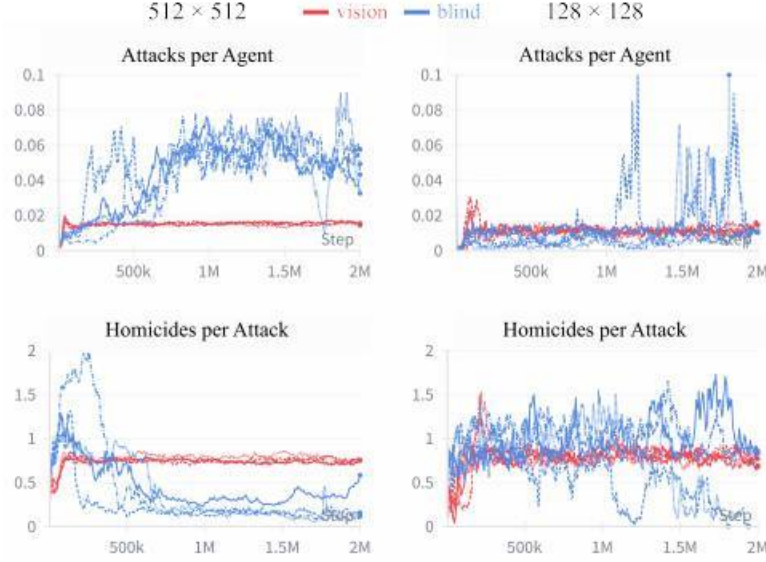


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in the 512x512 and 128x128 **Ocean** worlds. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agents attack less frequently but with much higher precision, pointing to the emergence of specialized predation behaviors. This dynamic grows more pronounced at larger scales. See Figure 10 in Appendix C for a comparison of grid sizes 1024x1024 and 256x256.

4.3 ABLATIONS

To assess how broadly our findings hold across configurations, Appendix F presents ablations over a range of hyperparameters, including initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8), measuring how sensitive the emergent behaviors are to simulator configuration. In addition, Appendix F.3 presents experiments verifying that the dynamics observed in larger environments are not simply the result of self-averaging.

5 CONCLUSION

Our study highlights the promise of open-ended ecological settings as a broadly applicable paradigm for investigating the emergence of intelligence. Across experiments spanning worlds of varying scale and complexity, we find that both environmental scale and sensory richness contribute substantially to the development of sophisticated sensing and decision-making policies. Taken together, these results suggest that ecological and evolutionary pressures, when coupled with sufficiently rich environments, may offer a natural and general pathway toward the development of increasingly intelligent artificial agents. Our experiments further demonstrate that rich behaviors can be evolved in large populations within a reasonable hardware budget and without resorting to explicit objectives. We hope these results will encourage wider exploration of ecology as a mechanism for developing and understanding artificial intelligence.

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A ADDITIONAL DATA

Table 3: Mining events in the **Beach** world for populations of agents with a compass (**RC**) compared to populations without a compass (**R**), spanning five scales.  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum marking the start of the mining event. For runs with multiple mining events, we report the first.  denotes the timestep at which the run went extinct, or n/a if it survived the full 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k			
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓			
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4			
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			
2	x			n/a	x			1070k	x			165k	x			55k	x			
3	x			n/a	x			280k	x			125k	x			90k	x			
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			

Table 4: Mining events across the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for **(RC)** agents compared to **(R)** agents. Grid size is 512×512 with an initial population of 8192.

Island					Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging data for Vision and Blind agents in the **Ocean** world across scales. denotes population size. denotes biomass utilization, the percentage of total biomass in the system currently held inside the agents' stomachs. denotes move actions per agent, the fraction of time an agent took a move action, averaged across agents. Similarly, denotes the proportion of agents' actions that were eat actions. Reported values of , , , and are averages of the metric over all steps for the seed.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation data for Vision and Blind agents in the **Ocean** world across scales.  denotes attacks per agent, the fraction of time an agent took the attack action, averaged across all agents.  denotes homicides per attack, the number of agents killed per attack action. Reported values of  and  are averages over all steps for the seed. Note the outlier of 0.562 for  in seed 3 of the 128×128 (**RC+A**) run: this seed went extinct before 2M steps, producing a spike in the attacks-per-agent metric as the population dwindled.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We compare rates of extinction and mining emergence as a function of world size and of agent access to global direction via the compass sensor. Extinction is defined as the population reaching zero individuals, while mining behavior is defined as in earlier experiments (a 10% drop in free biomass on land).

Runs are grouped by both world size and compass access. We simulated 64 runs each of 128×128 with no compass and 128×128 with compass, and 32 runs each of 512×512 with no compass and 512×512 with compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution (treating runs that did not go extinct within 2M steps as right-censored). As Figures 7 and 8 show, worlds tend to survive longer and to produce mining more quickly whenever they are large or whenever agents have compass information.

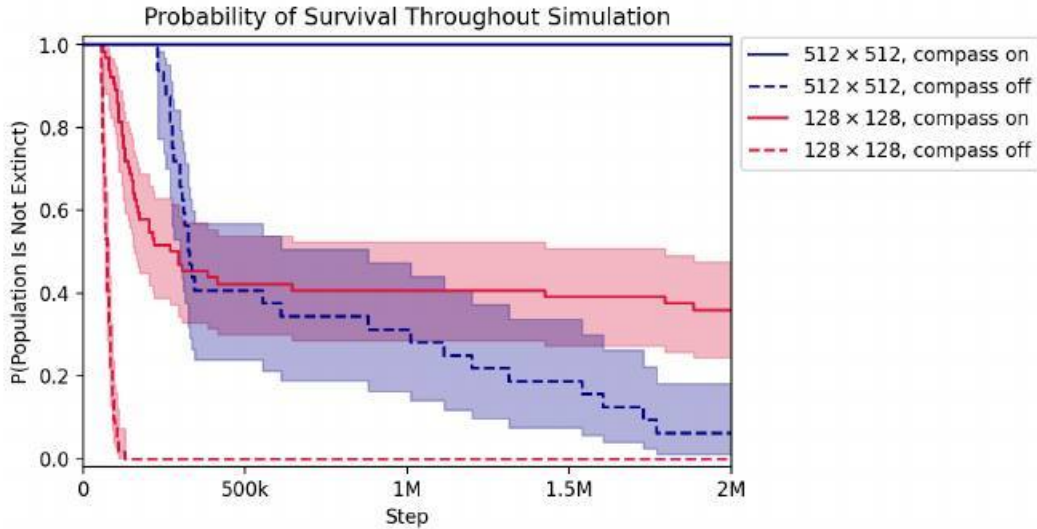


Figure 7: Probability that the population has not gone extinct as a function of world size, compass access, and timestep. We fit a separate non-parametric Kaplan–Meier estimator for each group of runs (e.g., 512×512 , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each 512×512 group fits the estimator on 32 runs; each 128×128 group fits on 64 runs.

We compare emergence and extinction timing with a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazard rates (Table 7). Runs that did not experience an emergence or extinction event by the end of 2M time steps were treated as censored. For extinction, both larger world size and compass use substantially reduce the hazard: the extinction rate at 128×128 is about 25 times greater than the corresponding rate in 512×512 worlds ($HR=0.04$), and enabling the compass produces an effect of similar magnitude ($HR=0.03$). For mining emergence, larger worlds generate mining behavior much more rapidly ($HR=6.07$), while compass use accelerates emergence dramatically ($HR=34.59$). The models show good discriminative ability between groups (concordance 0.80 for extinction, 0.85 for mining). Overall, larger worlds can sustain larger populations, reducing the chance of hitting the absorbing zero boundary while simultaneously exploring a larger phenotype space through the greater quantity of mutation available relative to smaller worlds.

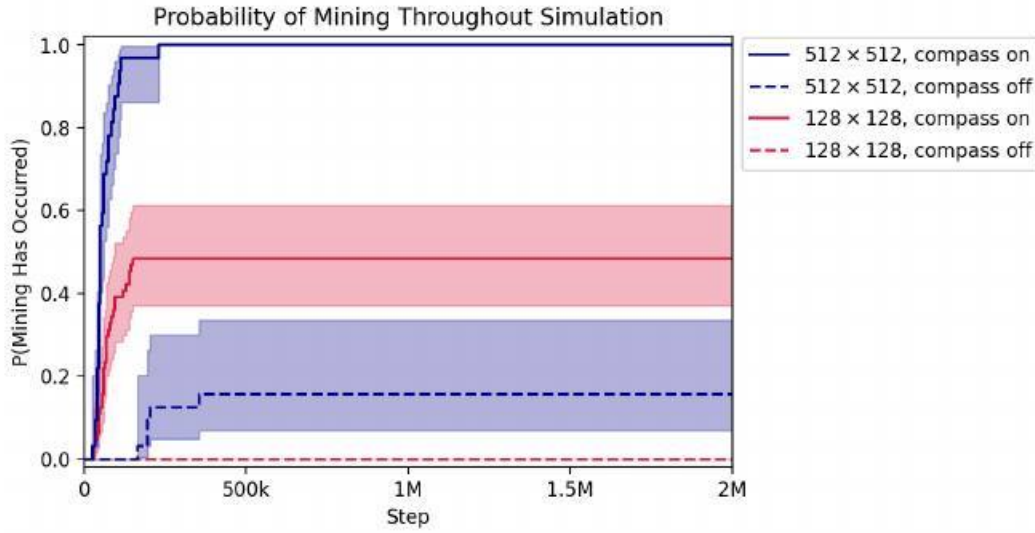


Figure 8: Probability that mining has evolved within the population as a function of world size, compass access, and timestep. We fit a separate non-parametric Kaplan-Meier estimator for each group of runs (e.g., 512×512 , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each 512×512 group fits the estimator on 32 runs; each 128×128 group fits on 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events, using a 128×128 , compass-off baseline. Hazard ratios (HR) below 1 indicate reduced hazard (a longer time to the event); for example, extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the rate of 128×128 . There are 192 runs in total: 32 seeds with compass on and 32 with compass off in the 512×512 world, and 64 seeds with compass on and 64 with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

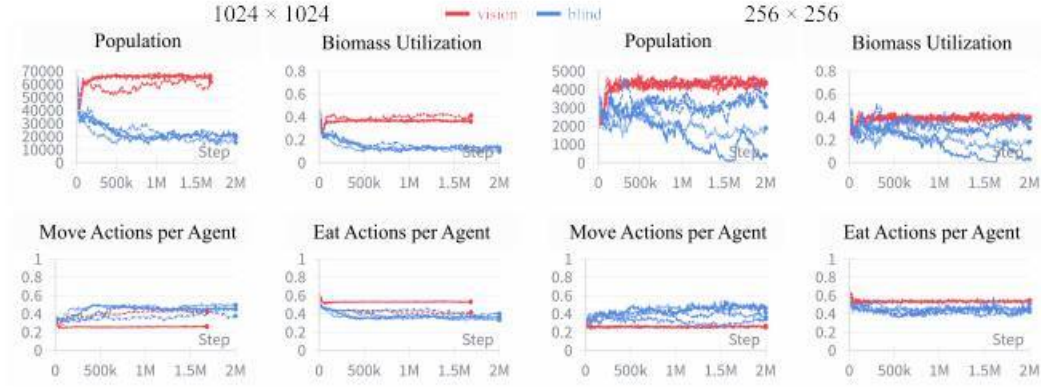


Figure 9: Foraging behaviors in the **Ocean** world across scales. Enabling vision in agents (red) yields more stable populations and greater biomass utilization, achieved through more frequent and effective eating and less movement. At larger scales these dynamics grow more stable, with less variance across runs. Note that, owing to limitations on GPU hours, the vision seeds at the 1024×1024 scale ran for 1.7M rather than 2M steps.

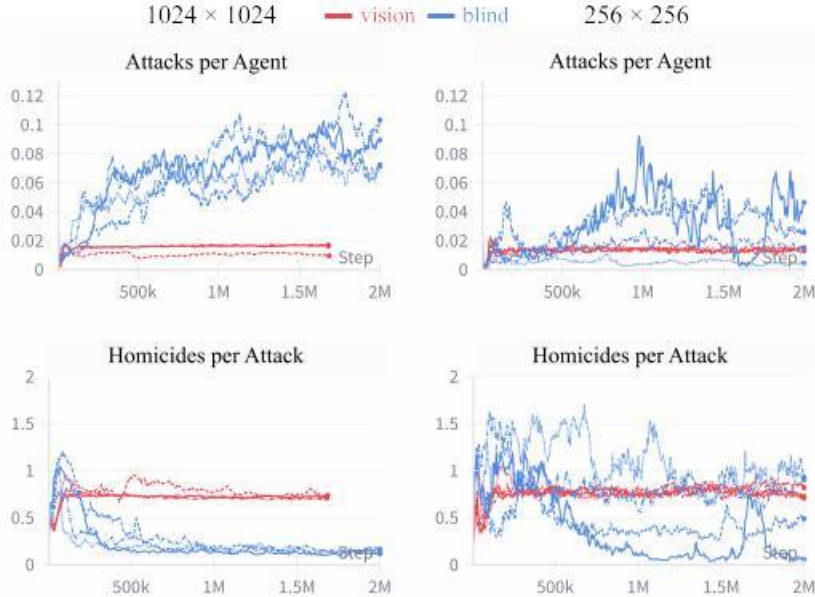


Figure 10: Predation behaviors in the **Ocean** world across scales. With vision (red), agents attack less frequently but with much higher precision, pointing to the emergence of specialized predation behaviors alongside foraging capabilities (Figure 9). This dynamic is again more pronounced at larger scales, with greater variance in the behavior of blind agents (blue) at smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

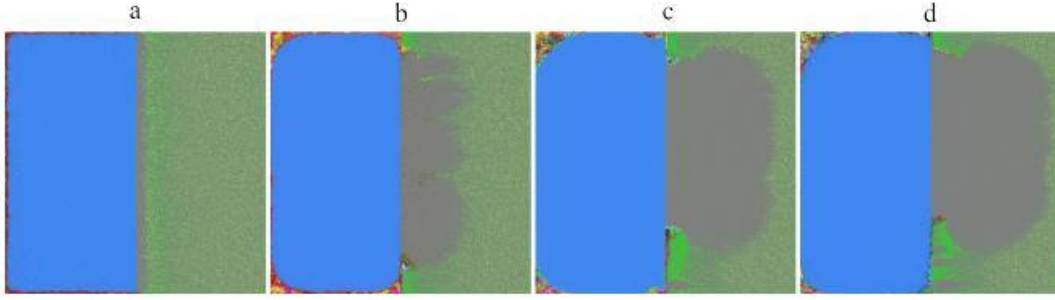


Figure 11: Larger-scale environments give rise to more varied and complex ecologies, producing distinctive visual patterns. In this 1024×1024 **Beach** world, **(RC)** agents initially died at the shore (a), transferring biomass to dry land. They then developed mining behavior, clearing biomass from the beach (b). When biomass re-accumulated at the ends of the beach (c), subpopulations of agents independently re-developed mining behavior, traveling inland along these smaller biomass beaches to extract the resources (d).

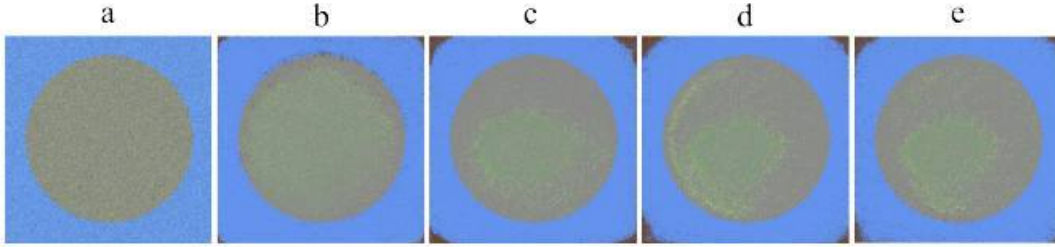


Figure 12: Mining behavior emerges across a variety of terrains. In this 512×512 **Island** world with **(RC)** agents, different regions of the shore are mined over the course of the run (b) (c). Biomass is transferred back onto parts of the island once mining becomes difficult and agents die before they can return to water (d). Agents then adapt to re-mine this wall of accumulated biomass, clearing it from the shore (e).

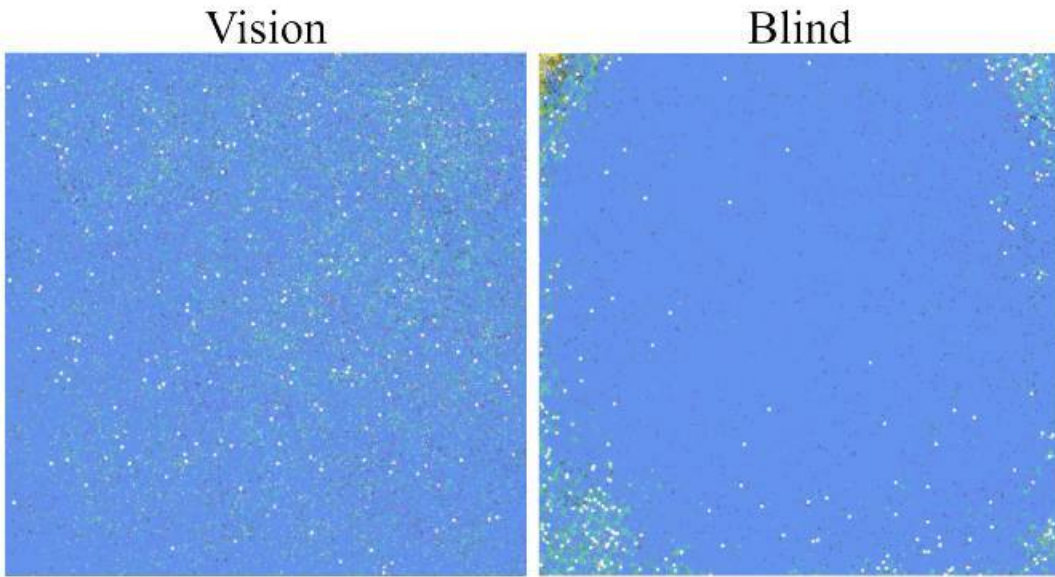


Figure 13: Representative runs in a 512×512 **Ocean** world with attacking enabled: vision **(RCV+A)** agents on the left and blind **(RC+A)** agents on the right. White squares indicate that an agent is attacking at that location. With vision, agents adapt to locate free resources throughout the ocean world and to prey on other agents more effectively (see Figures 5 and 6). Blind agents explore the ocean for resources far less, instead adapting policies for clustering in the corners of the world.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We report details on simulation design, runtime, and experiment reproducibility.

Simulator design. All experiments are conducted using a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. The simulator computes every part of the environment step on the GPU—terrain dynamics, resource flows, agent updates, and measurement—so that policy evaluation and simulation share a single JAX compilation pipeline. It exposes a minimal population-level interface: given the current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation lets the same environment implementation be reused with a broad range of population update rules, including evolutionary algorithms, reinforcement learning methods, and others. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This yields the throughput required for our scaling experiments, with runtime measurements for a single H100 GPU reported in Table 8.

A key feature of the simulator is its built-in tooling. A measurement API computes and records aggregate statistics—such as biomass utilization, movement and eating rates, attack and homicide rates, and a broad set of environment-level parameters—at every step, which we use directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from generated reports and supports inspection of individual agents (Figure 1). These tools were critical for discovering and validating qualitative phenomena, such as long-range resource mining and predation patterns, that are not apparent from scalar metrics alone.

From a high-performance computing perspective, our simulator fills a gap left by existing systems. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), and Griddly (Bamford et al., 2020) offer broad configurability but require substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study. GPU-native multi-agent RL environments, conversely, emphasize fixed numbers of agents and task-specific reward structures. Our simulator instead combines GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over 10^5 agents while remaining interoperable with a wide range of population algorithms. This combination yields a state-of-the-art platform for scaling ecological experiments like those in this paper, as well as a general foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 reports runtime data across a range of grid sizes and maximum agent populations. Each run was conducted on a single H100 GPU for a maximum of 2M steps. While the table reports run-times for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total), runtime varies with the size of the agent networks, the sensors available, and the level of logging.

Our largest runs, at the 1024×1024 grid size, take ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on one H100, with memory usage of roughly 60GB. **Ocean** runs are slightly slower owing to the enabled agent attack action and vision sensing.

Table 8: Runtime data in the **Beach** and **Ocean** worlds across a range of grid sizes and maximum agent populations. Each (RC) agent has a 22,465 parameter neural network, and each (RCV + A) agent has a 22,529 parameter neural network. We report simulation steps per second (hz) and the total extrapolated wall-clock runtime in hours for 2M steps. Effective runtime is lower for seeds in which populations go extinct before 2M steps.

Environment	1024×1024		512×512		256×256		128×128		64×64	
	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

Reproducibility. Because of the large and randomized nature of our environments, the precise outcome of an experiment can differ substantially from run to run. For this reason, we report results across four distinct random seeds for each experiment. Owing to nondeterminism in some low-level GPU operations, we also find that repeating an experiment with the same random seed does not reliably reproduce the same results. In particular, modern GPUs may execute parallel floating-point operations such as summations in different orders across runs, and may fuse operations in ways that alter rounding behavior. Very small differences arising from finite-precision arithmetic can accumulate over many steps into noticeable drift in the simulation trajectory. The use of 16-bit weights can coarsen these rounding effects but is not their primary source. Despite this run-to-run variation, clear high-level patterns emerge consistently once the results of multiple seeds are taken in aggregate.

F ABLATIONS

We present a series of ablations probing how specific hyperparameters affect our results and how robustly those results hold across different environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate how initial population size affects the consistency of mining behavior and population stability in a 256×256 **Beach** world. Results are reported for **(RC)** agent populations. Agents without a compass **(R)** exhibited a 0% mining rate and 100% extinction rate across all initial population sizes except 8192, the highest tested, which reached 25% mining and 75% extinction frequency. Note that mining and extinction frequencies need not sum to 100%: a population may, for instance, develop mining and later go extinct, or persist for the full 2M steps of our experiments without ever developing mining behavior.

As shown in Figure 14, initial population size has no significant effect on mining consistency or extinction rates: populations starting with only 128 agents evolve mining behaviors as often as populations starting with 8192 agents, and extinction rates remain largely stable across the full range of initial population sizes.

Figure 15 offers insight into why so little effect is visible. Regardless of the initial population size, each run rapidly converges to a similar population size within the first 50,000 steps of simulation. This suggests that the environment implicitly specifies a carrying capacity determined by the fixed resources available in the system: populations beginning below the carrying capacity grow to reach it, while populations that are too large are pulled back down toward it. As a result, the effective population size is similar once the initial population size has been corrected early on, leaving mining and extinction frequencies unaffected. Larger initial populations do, however, contribute somewhat to the total resources in the system, since agents carry biomass that is deposited on death; consequently, larger initial population sizes reach higher maximum populations on average over the course of simulation (Table 9), though the effect is not strong enough to alter mining or extinction frequency in this setting. It is worth noting that although the population size converges early, it does not remain constant throughout the run. Later in the simulation, populations across different runs can diverge substantially, as the ecological consequences of agents evolving different abilities at different times reshape the later resource distributions.

F.2 VARYING RESOURCE RULES

We evaluate how two resource hyperparameters—the biomass photosynthesis rate and the initial biomass site density—affect mining and extinction frequency. The setting matches that of Appendix F.1. The initial population size is fixed at 2048 for this study, and agents once again have a compass **(RC)**.

The biomass photosynthesis rate controls how quickly landscape biomass creates energy via photosynthesis, supplying resources to agents over time. The initial biomass site density parameter sets the fraction of grid cells containing biomass at the start of simulation. All experiments in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

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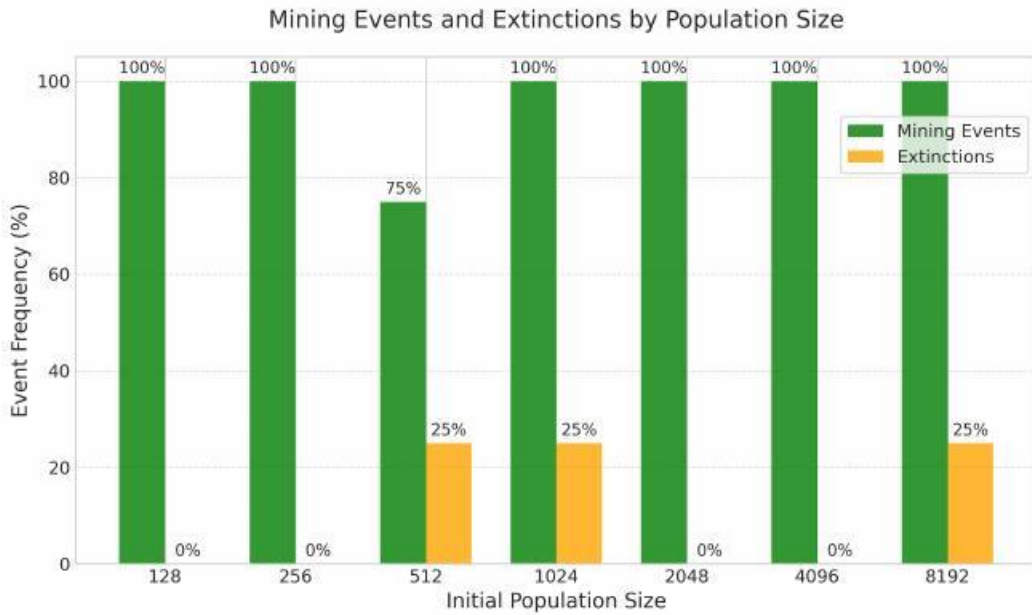


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world across a range of initial sizes of (**RC**) agent populations. Frequencies are computed over four random seeds for each initial population size; e.g., a 25% Extinction Event Frequency means that in 1/4 seeds the population reached 0 before 2M simulation steps.

Table 9: The maximum population size reached after step 0 across a range of starting population sizes, with averages and standard deviations over four random seeds. Larger initial population sizes reach slightly higher maximum populations, with greater variance across seeds than smaller initial population sizes. Overall, however, the maximum population reached is largely robust to the initial population size: an initial population of 128, only 1.6% of an initial size of 8192, still reaches 78.9% of the maximum population reached by the latter (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

Whereas varying the initial population size left mining and extinction consistency unchanged (Figure 14), varying the resource rules has a pronounced effect, as shown in Figures 16 and 17.

Intuitively, larger biomass photosynthesis rates or initial biomass site densities directly raise the effective carrying capacity of the environment by increasing the resources available in the system. This in turn sustains larger populations throughout the run, increasing the likelihood of mutations favoring mining behavior while lowering the chance of extinction. Our data bear out this intuition: on average (omitting seeds that went extinct), runs with initial biomass site density 1/16 reached a maximum population of 1608 after step 0, rising to 2466 for density 1/8, 3786 for density 1/4, and 5353 for density 1/2.

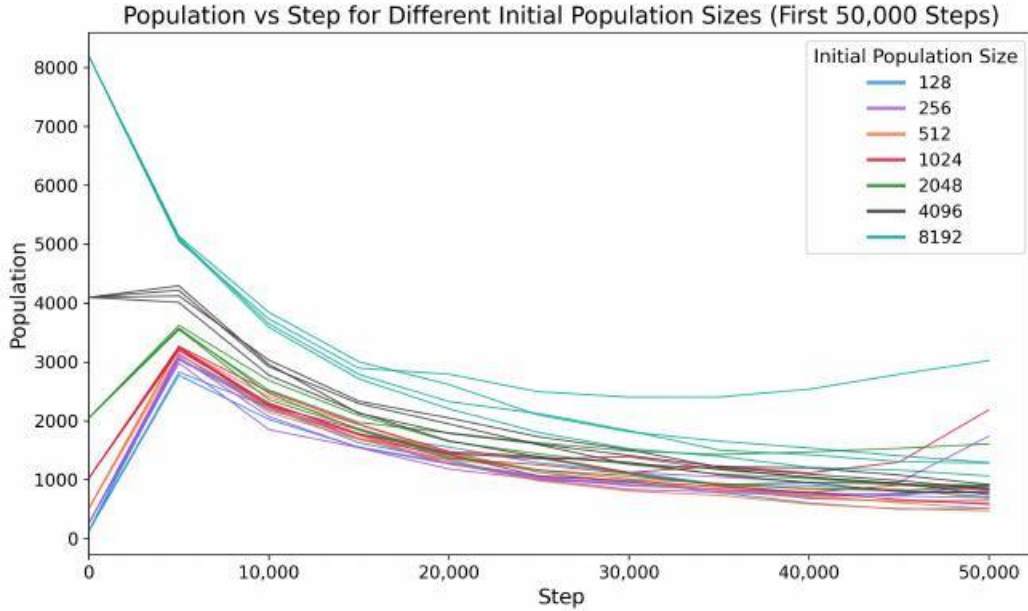


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with (RC) agents, with four seeds per initial population size. The environment supports an implicit carrying capacity set by the resources available in the system, and all initial population sizes converge toward this value early in the run. Larger initial population sizes contribute to the total resources in the system through biomass deposited on agent death, raising the carrying capacity somewhat.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One natural concern is that the increased stability of mining behavior in large environments (Table 1) might be explained purely by self-averaging rather than by qualitatively different dynamics. By self-averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, so that a single large system behaves like the average of smaller systems of the same total size. On this view, a 512×512 **Beach** world might simply behave like many loosely coupled 128×128 worlds run in parallel. Were that the case, our scaling results could in principle be reproduced by running enough small worlds and averaging over random seeds.

To test this hypothesis directly, we run an experiment that holds the total “sample size” roughly fixed while varying how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (RC), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1. As a control, we repeat the entire protocol for agents without a compass (R). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

If large environments were acting as self-averages of many smaller ones, we would expect mining and extinction rates to be roughly invariant to how grid cells and agents are distributed across runs:

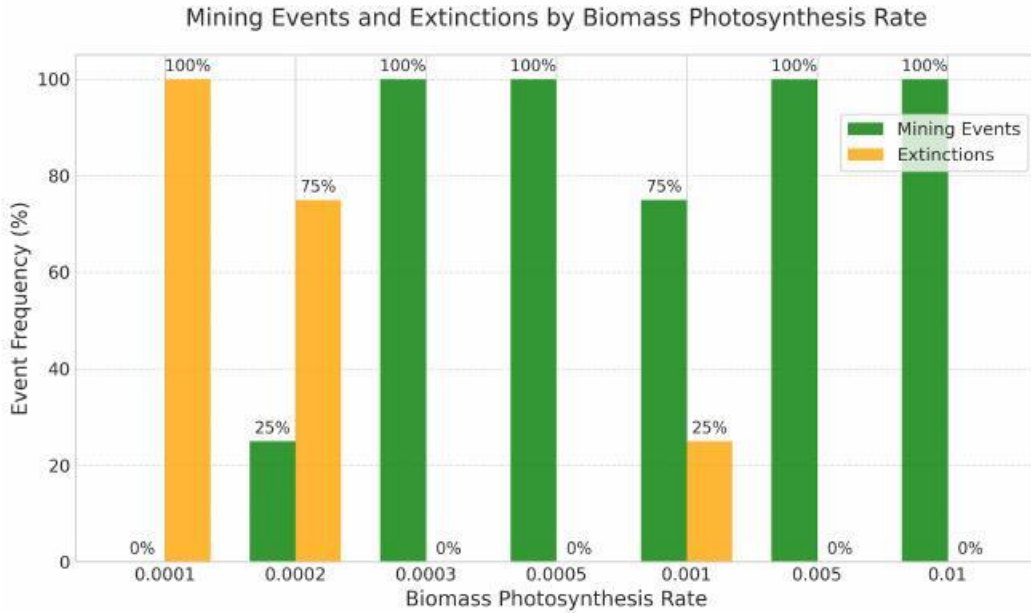


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world across a range of settings of the biomass photosynthesis rate resource parameter. Frequencies are computed over four random seeds for each setting. Mining consistently emerges at biomass photosynthesis rates of 0.0003 and greater.

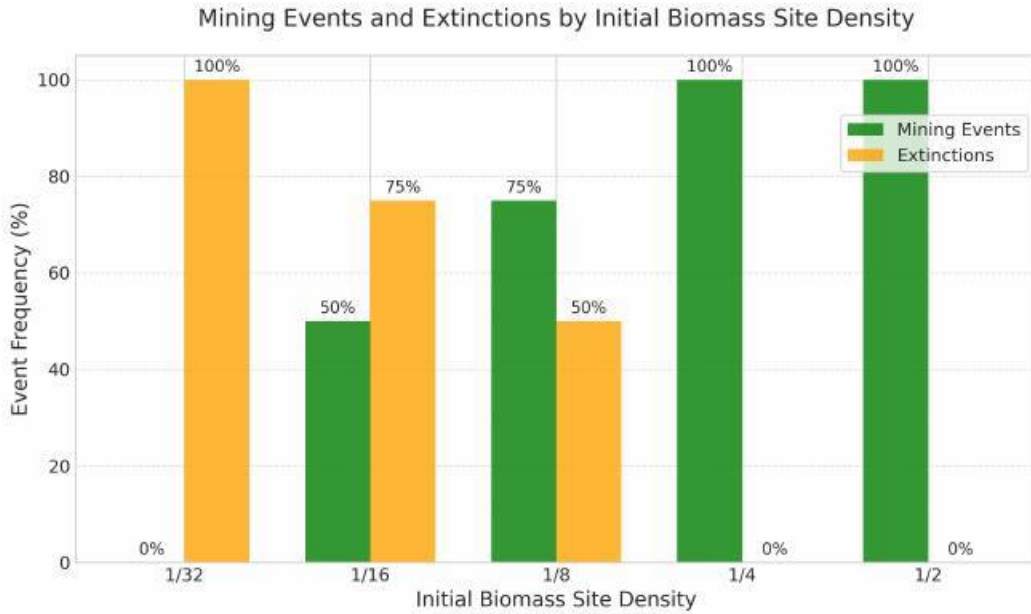


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world across a range of settings of the initial biomass site density resource parameter. Frequencies are computed over four random seeds for each setting. Greater initial biomass site densities yield more consistent discovery of mining and a lower chance of extinction.

4 large worlds or 64 small ones would exhibit similar probabilities of discovering mining. Instead, we observe a strong dependence on world size. For (RC) agents, mining emerges in all runs with no extinctions at 512×512 , remains common but not guaranteed at 256×256 , and becomes unreliable

and fragile at 128×128 , where many populations go extinct before reaching 2M steps. In contrast, **(R)** agents never develop mining and go extinct at every scale, confirming that the compass remains a key enabler of long-range behavior throughout.

Table 10: Mining and extinction events when trading off world size against the number of independent runs. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , and 128×128 , with initial populations scaled proportionally (8192, 2048, and 512 agents, respectively). For each size, we allocate the same total number of grid cells and initial agents across all runs, using 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged (🔧) and the fraction in which the population went extinct before 2M steps (💀), for agents with a compass (**(RC)**) and without a compass (**(R)**).

	512×512 8192		256×256 2048		128×128 512	
	🔧	💀	🔧	💀	🔧	💀
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These results are inconsistent with a self-averaging explanation. On the contrary, larger environments appear to enable and stabilize mining behavior in ways that cannot be accounted for by averaging over more small simulations alone.

Even when the total number of cells, agents, and simulation steps is matched across conditions, concentrating resources into a few large worlds produces a regime in which mining is consistent and extinctions are rare, whereas dispersing the same resources across many small worlds yields populations that frequently fail to discover mining and are more prone to collapse. This suggests that increasing environmental scale does not merely smooth out stochastic noise, but instead alters the underlying ecological dynamics—for example, by supporting larger, more spatially distributed populations and more persistent diversity among evolving policies.

F.4 MUTATION RATE SWEEP

We test how sensitive the emergence of mining is to the mutation rate hyperparameter, which controls the scale of Gaussian noise added to a child agent’s policy upon birth. Intuitively, the mutation rate governs how different a child agent is from its parent. We once again run experiments in a 256×256 **Beach** world with an initial population of 2048 **(RC)** agents for a maximum of 2M steps.

As Figure 18 shows, mining emerges in at least 3 of the 4 seeds across the entire range of mutation rates from 6×10^{-3} to 7.875×10^{-2} , indicating that our Section 4.1 results with mutation rate 3×10^{-2} are robust throughout this interval. As expected, mutation rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, leading to extinction, while overly large rates (e.g., 3×10^{-1}) undermine population stability, consistently causing extinction with no mining adaptation.

F.5 NETWORK SIZE

We study how the size and architecture of the agent policy network affect mining and extinction frequency, and how environmental scale in turn modulates this effect. We investigate 256×256 and 512×512 **Beach** worlds with initial population sizes of 2048 and 8192 **(RC)** agents, respectively (scaling by four with the number of grid cells). For each experiment we also ran baselines with **(R)** agents, again finding rare to no mining events and 100% extinction rates.

The main experiments in Section 4 used agent policies with 2 layers and 64 channels, yielding a network of 22465 parameters per agent (Table 11). As Figure 19 shows, in the 256×256 **Beach**, halving the channels to 32 (9185 total parameters) or doubling them to 128 (61313 parameters) while holding the number of layers fixed has minimal to no effect on mining and extinction frequency. Changing network depth, by contrast, has a substantial effect: all populations of agents with just

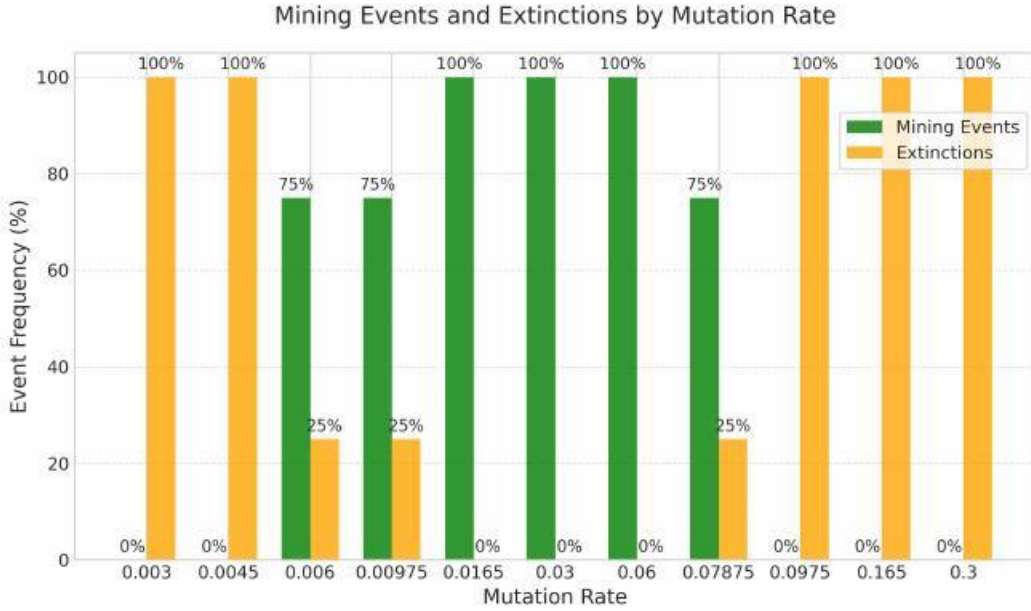


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world across a range of mutation rates. Frequencies are computed over four random seeds for each setting.

1 layer and 32 channels went extinct, as did 50% of populations of agents with 3 layers and 128 channels.

Crucially, scaling the environment to the 512×512 grid size produced markedly different dynamics for networks of varying depth (Figure 20). In the larger world, agent populations with 3-layer, 128-channel networks consistently mined and survived through 2M steps, and, more surprisingly, populations with the small 1-layer, 32-channel networks mined consistently as well. We hypothesize that both smaller and larger networks find it harder to discover new behaviors, though for different reasons. When the network is smaller, its reduced capacity makes good solutions hard to find; when the network is larger, the expanded search space means promising new behavior takes longer to reach. Regardless of the precise cause, the fact that these networks fail so often in small environments yet succeed in larger ones further illustrates the general stabilizing benefit of increasing scale. A larger environment can support larger populations, allowing for more exploration in policy space, and at the same time it takes longer for locally successful mutants to propagate across a large environment, which promotes diversity and again ensures greater exploration. For further discussion of these effects, see (Mitchell et al., 2006).

Table 11: Total parameters in (RC) and (R) agent policies across a range of network architectures.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

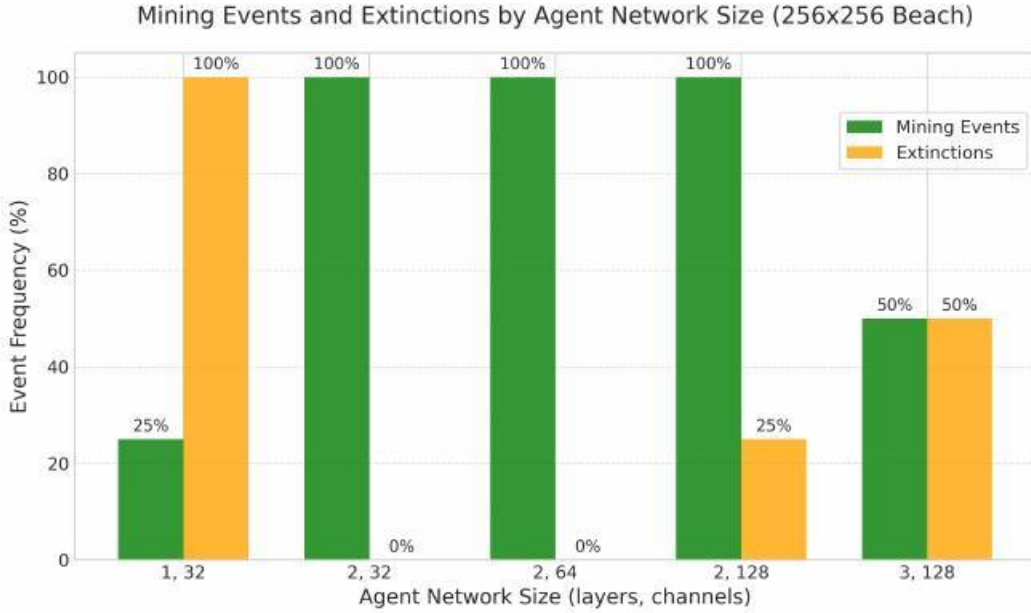


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world across a range of agent policy network sizes. Frequencies are computed over four random seeds for each setting.

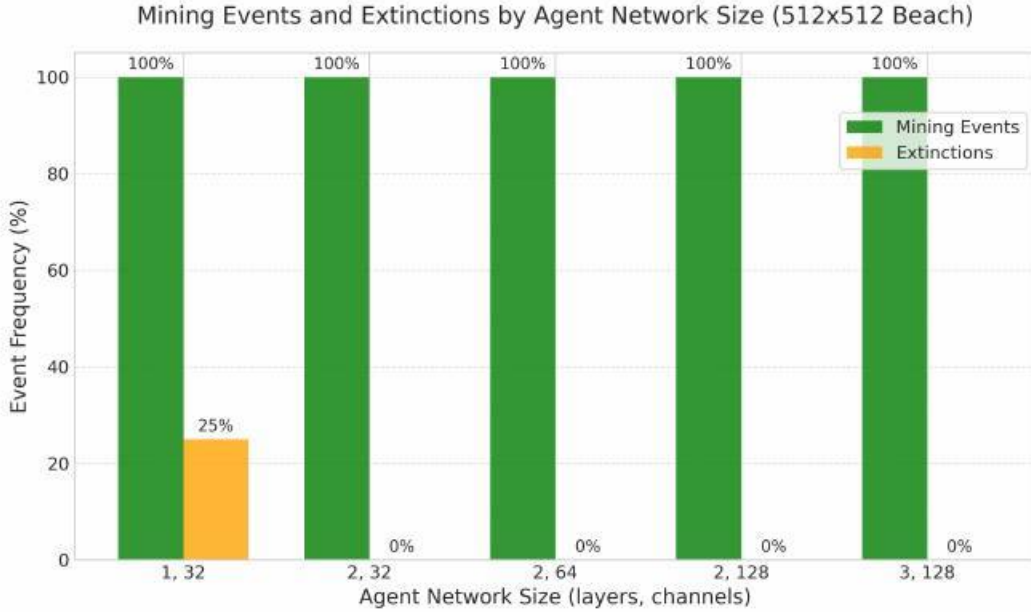


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world across a range of agent policy network sizes. Frequencies are computed over four random seeds for each setting.

F.6 VARYING VISION RANGE

We assess how different vision ranges and viewing configurations shape predation and foraging behaviors in the 512×512 **Ocean** world with (RCV+A) agents. We evaluate visual sensor ranges of 3×3 , 7×7 , and 9×9 , and for each range we consider the two viewing configurations shown in Figure 21 a) Around, in which agents can see grid cells all around them, and b) Front, in which agents cannot see cells behind them. The total number of cells in the agent’s vision range is held

equal across the two configurations. The main experiments with vision in Section 4.2 were run with the 7×7 Around setting.

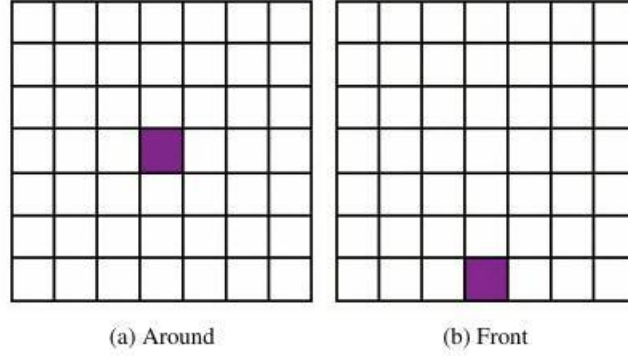


Figure 21: Comparison of the Around and Front viewing configurations, illustrated for the 7×7 vision range. The purple cell represents an agent, facing the top of the page.

Table 12: Foraging and predation data in the 512×512 **Ocean** world for vision agents (**RCV+A**) across a range of vision ranges and viewing configurations. Refer to Tables 5 and 6 for definitions of the metrics represented by each icon. For each seed, every metric was averaged over all steps starting at step 500k to allow the values to stabilize. Reported values are the mean across the four seeds together with the standard deviation of the four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

These results indicate that a smaller 3×3 vision range yields behavior largely similar to no vision at all (Blind), while the 9×9 results closely track those at 7×7 . This points to diminishing returns from larger vision ranges in the environmental configurations we study. Such diminishing returns are intuitive: since the agents have no memory, it is unlikely that visual features at the periphery of a 9×9 window could meaningfully alter the optimal action relative to a 7×7 window consistently enough to provide a strong learning signal.

The data also suggest that at the 7×7 and 9×9 vision ranges, the Around configuration is slightly more favorable in terms of population stability and the effectiveness of attack and eat actions, yielding higher average population , biomass utilization , and homicides per attack  than the Front configuration. We again attribute this to diminishing returns from seeing more distant pixels: in the Around configuration, more of the pixels in the agent’s vision range are nearby and thus likely carry more relevant information for near-term actions than the distant cells visible under the Front configuration. At the 3×3 vision range, however, the trend reverses, with Front more favorable than Around; this indicates that here the marginal benefit of seeing one more pixel ahead under the 3×3 Front configuration outweighs that of seeing one pixel behind under the 3×3 Around configuration, whose behavior is most similar to the Blind setting.

F.7 VARYING ATTACK STRENGTH

We study how reducing agent attack strength shapes emergent behaviors and population dynamics in the 256×256 **Ocean** world. We evaluate three attack strengths: 1) Full Attack, dealing 10 HP of damage and instantly killing any agent in the attack radius; 2) Half Attack, dealing 5 HP of damage;

and 3) Quarter Attack, dealing 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 were run with the Full Attack setting.

Reducing the attack strength from Full Attack to Half Attack substantially lowers both the effectiveness and the frequency of attacks: attacks per agent and homicides per attack both drop considerably (Figure 22). As a result, populations with lower attack strength grow larger, with agents moving and eating more frequently and thereby achieving greater biomass utilization. Further reducing strength to the Quarter Attack setting yields diminishing returns, with little additional reduction in violence or growth in population. Intuitively, when attacks do less damage, each homicide requires more attacks, reducing the lethality of any individual attack. As lethality falls, attacking becomes less useful and therefore less likely to be selected for in the population. Agents with lower attack strengths thus become less skillful at attacking and spend more time foraging for food.

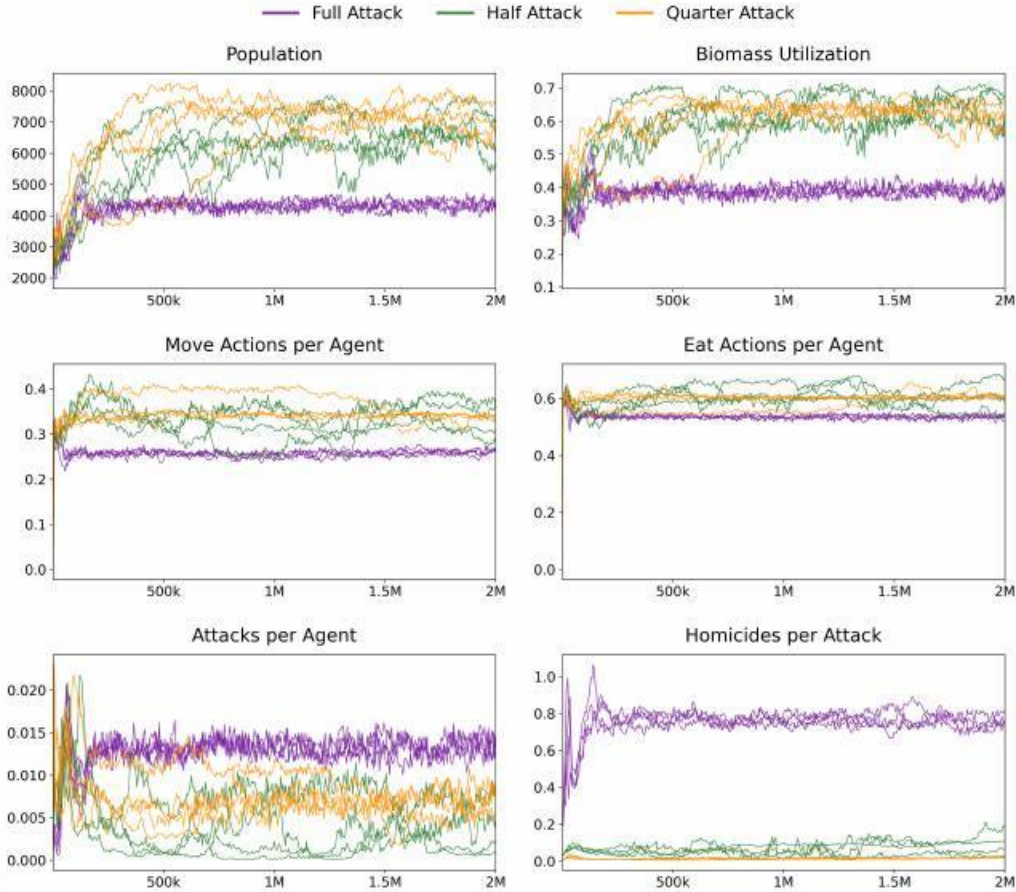


Figure 22: Population and action frequency data over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents across the Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted per attack strength setting.

F.8 VARYING AGENT HP

We investigate how reducing the starting HP of child agents and lowering the healing rate shape behavior, applying pressure for agents to ‘grow’ over their lifetime in order to survive. The healing rate parameter determines how much HP an agent regenerates per healing step, which converts available energy into HP. Agents can influence healing only by maintaining enough energy.

We study two settings: 1) Full HP, in which child HP starts at 5 and the healing rate is 1 HP per 0.1 energy; and 2) Low HP, in which child HP starts at 1, with healing rate 0.5. Maximum agent HP is 10 in all settings. The main experiments (Section 4) were run with the Full HP setting.

Across the two settings, there is no significant difference in homicides per attack, eat action frequency, or biomass utilization (Figure 23). Under Full HP, populations grow slightly larger, while under Low HP, agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must concentrate on foraging early in life in order to grow to full size, which would account for this difference.

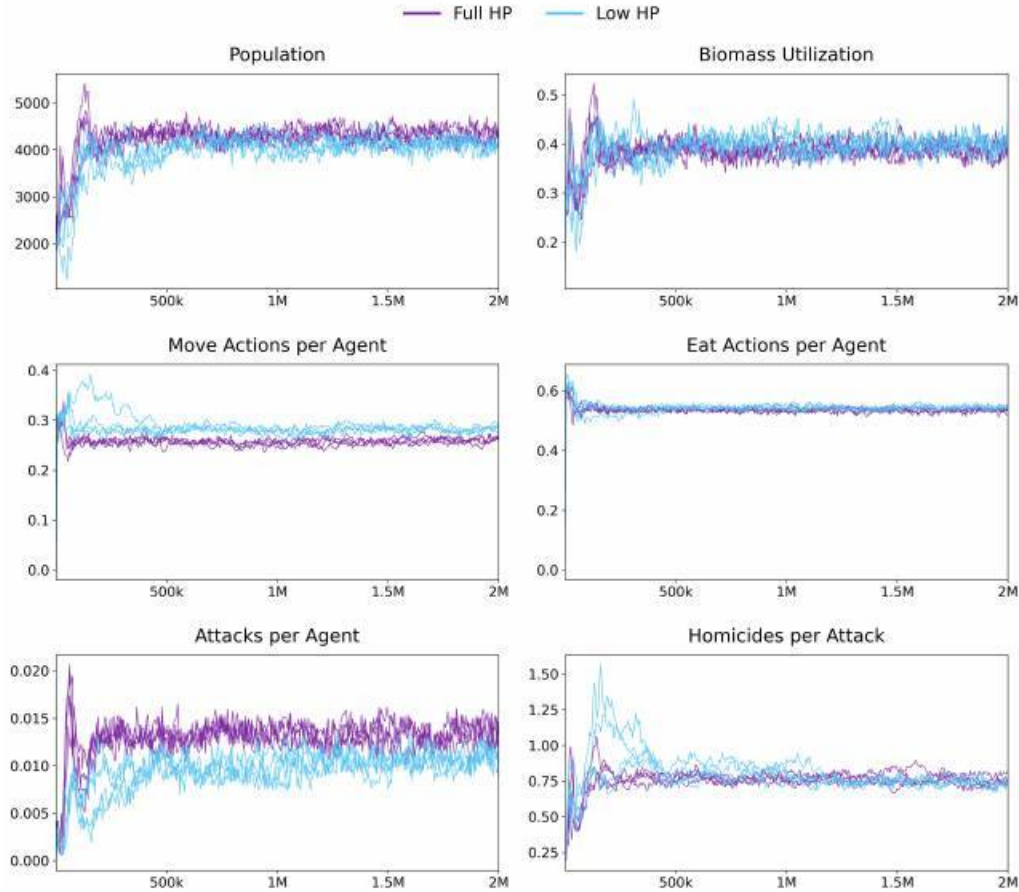


Figure 23: Population and action frequency data over 2M steps in the 256×256 **Ocean** world for **(RCV+A)** agents across the Full HP and Low HP settings. Four seeds are plotted per HP setting.